

CPTG and the Hubble-Tension Bridge

A Native-Geometric Resolution Through Acoustic and Luminosity Projections

Carter L Glass Jr

Abstract

The Hubble tension compares an early-universe CMB/acoustic inference of the present expansion scale with a locally calibrated distance-ladder inference. Curvature Polarization Transport Gravity (CPTG) organizes these observations through one native cosmological π branch and two fixed comparison projections, acoustic and affine luminosity. The three descriptions share the same geometric phase $N = \ln a$ but accumulate branch-specific clock readings through $H_b dt_b = dN$.

The native CPTG normalization and exact acoustic projection give

$$H_{0,\text{ac}}^{\text{CPTG}} = 67.4777967351 \text{ km s}^{-1} \text{ Mpc}^{-1},$$

which differs from the Planck 2018 base- Λ CDM CMB inference $67.4 \pm 0.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$ by 0.156σ when the CPTG coordinate is treated as fixed. Independently, the registered affine luminosity normalization is

$$\mathcal{G}_L^{\text{aff}} = \frac{10}{9}, \quad \Gamma_L^{\text{aff}} = \sqrt{\frac{10}{9}}, \quad \Xi_L^{\text{aff}} = \sqrt{\frac{9}{10}},$$

which gives

$$H_{0,L}^{\text{aff}} = 73.1711530416 \text{ km s}^{-1} \text{ Mpc}^{-1}.$$

The four-anchor SH0ES result $73.17 \pm 0.86 \text{ km s}^{-1} \text{ Mpc}^{-1}$ is therefore an external comparison rather than an input to the luminosity construction. The theory endpoint differs from its central value by only $0.0011530416 \text{ km s}^{-1} \text{ Mpc}^{-1}$, or approximately 0.00134σ . The acoustic and affine luminosity endpoints are separated by 8.43737730% . The bridge therefore represents the Planck and SH0ES endpoints as external tests of two fixed projections of one native geometric history rather than using either observation to define the native branch or the opposite projection.

1 Introduction

The Hubble tension is one of the most persistent consistency problems in modern cosmology. Measurements tied to the early Universe, most prominently the cosmic microwave background, favor a present expansion scale near $67 \text{ km s}^{-1} \text{ Mpc}^{-1}$, while locally calibrated distance ladders favor a value near $73 \text{ km s}^{-1} \text{ Mpc}^{-1}$. The difference is much larger than the quoted observational uncertainties and has survived major improvements in both early- and late-Universe measurements. Within a framework in which these determinations are required to represent the same present-day expansion parameter, the discrepancy is difficult to accommodate: either one or more measurement pipelines contain an unresolved systematic error, or the cosmological model used to connect early- and late-Universe observations is incomplete. For that reason the Hubble tension is not simply a disagreement between two measurements; it is a test of whether one cosmological description can consistently connect the physical information carried by very different observational channels [1, 2, 3].

CPTG approaches the problem differently. It does not begin by requiring the CMB/acoustic and luminosity-distance measurements to return one identical observational H_0 coordinate. Instead, CPTG begins with a single native geometric π branch and constructs the acoustic and luminosity quantities as distinct projections of that same underlying history. The projections share the same native geometric phase but need not share the same projected rate or accumulated branch clock. In the present construction the native branch gives $H_0^{(\pi)} = 69.4162507897 \text{ km s}^{-1} \text{ Mpc}^{-1}$; its independently derived acoustic projection gives $H_{0,\text{ac}}^{\text{CPTG}} = 67.4777967351 \text{ km s}^{-1} \text{ Mpc}^{-1}$, while the independently derived affine luminosity projection gives $H_{0,L}^{\text{aff}} = 73.1711530416 \text{ km s}^{-1} \text{ Mpc}^{-1}$. The latter follows from the fixed CPTG normalization $\mathcal{G}_L^{\text{aff}} = 10/9$, not from the SH0ES result. Planck and SH0ES therefore enter afterward as external observational tests of two different projections rather than as competing estimates used to determine the native branch. In CPTG, what appears as a tension when both channels are forced onto one comparison coordinate is instead interpreted as a projection difference within one geometric cosmological history [4, 5, 6].

Quantitatively, a representative early-universe value is the Planck 2018 base- Λ CDM CMB inference

$$H_0^{\text{Planck}} = 67.4 \pm 0.5 \text{ km s}^{-1} \text{ Mpc}^{-1}, \quad (1)$$

while the four-anchor SH0ES distance ladder gives

$$H_0^{\text{SH0ES}} = 73.17 \pm 0.86 \text{ km s}^{-1} \text{ Mpc}^{-1} \quad (2)$$

[1, 2]. The SH0ES value combines four geometric distance anchors with homogeneous HST Cepheid photometry and the Cepheid–SN Ia distance ladder. The underlying ladder is a simultaneous three-rung construction linking geometric Cepheid calibration, Cepheids and SNe Ia in nearby hosts, and SNe Ia in the Hubble flow [3].

The direct observational separation is

$$\Delta H_0^{\text{obs}} = 73.17 - 67.4 = 5.77 \text{ km s}^{-1} \text{ Mpc}^{-1}, \quad (3)$$

with combined quoted uncertainty

$$\sigma_{\text{obs}} = \sqrt{0.5^2 + 0.86^2} \simeq 0.995 \text{ km s}^{-1} \text{ Mpc}^{-1}. \quad (4)$$

Therefore,

$$\frac{\Delta H_0^{\text{obs}}}{\sigma_{\text{obs}}} \simeq 5.80\sigma. \quad (5)$$

This arithmetic treats the two quoted uncertainties as independent and combines them in quadrature. It is distinct from a theory-to-observation residual in which a CPTG comparison coordinate is treated as fixed, and it does not remove the base- Λ CDM model dependence of the Planck inference.

The bridge developed below makes this distinction explicit: the observed Planck–SH0ES separation is retained as an empirical fact, while the CPTG acoustic and affine luminosity coordinates are derived before either observation is used as a test.

The central present-epoch relation is

$$H_{0,\text{ac}}^{\text{CPTG}} \longleftarrow H_0^{(\pi)} \longrightarrow H_{0,L}^{\text{aff}}. \quad (6)$$

Both projected endpoints are fixed before comparison with the corresponding observations. The more general branch structure is established next: one native geometric history is evaluated at a common phase while its acoustic and luminosity projections accumulate distinct branch times.

2 Native Branch, Projected Clocks, and Analytic Dependency Chain

CPTG uses a shared geometric phase

$$N \equiv \ln a. \quad (7)$$

For

$$b \in \{\pi, \text{ac}, L\}, \quad (8)$$

each branch clock is defined by

$$\frac{dN}{dt_b} = H_b(N), \quad H_b(N) dt_b = dN. \quad (9)$$

Thus equal N means equal geometric phase, not equal accumulated clock reading:

$$t_b(N) \not\equiv t_{b'}(N), \quad b \neq b'. \quad (10)$$

A luminosity comparison projector may be written generally as

$$H_L(N) = \Gamma_L(N) H_\pi(N), \quad \Xi_L(N) = \Gamma_L(N)^{-1}, \quad (11)$$

so that

$$\frac{dt_L}{dt_\pi} = \Xi_L(N), \quad H_L dt_L = H_\pi dt_\pi = dN. \quad (12)$$

For the registered affine luminosity projection used in this paper,

$$\Gamma_L(N) = \Gamma_L^{\text{aff}} = \sqrt{\frac{10}{9}}, \quad \Xi_L(N) = \Xi_L^{\text{aff}} = \sqrt{\frac{9}{10}}, \quad (13)$$

independent of phase. These factors are theory-side comparison coordinates and are not set by the SH0ES endpoint [5].

The native normalization used here is not selected by interpolating between the Planck and SH0ES values. It follows from the fixed CPTG geometric branch [4, 5]. The branch exponent and geometric acceleration anchor are

$$p_C = \pi, \quad (14)$$

$$a_{\pi,0} = \pi^{3/2} \times 10^{-10} \text{ m s}^{-2} = 5.568327996831708 \times 10^{-10} \text{ m s}^{-2}. \quad (15)$$

The late and transport weights are

$$\mathcal{B}_\pi = 1 - \frac{1}{\pi}, \quad \mathcal{T}_\pi = \frac{1}{\pi}. \quad (16)$$

The asymptotic expansion scale follows directly from the acceleration anchor,

$$H_\infty = \frac{a_{\pi,0}}{c} = 57.3131992073 \text{ km s}^{-1} \text{ Mpc}^{-1}, \quad (17)$$

and the present native normalization is

$$H_0^{(\pi)} = \frac{H_\infty}{\sqrt{\mathcal{B}_\pi}} = 69.4162507897 \text{ km s}^{-1} \text{ Mpc}^{-1}. \quad (18)$$

The native expansion law is therefore

$$E_\pi(a)^2 = \mathcal{B}_\pi + \mathcal{T}_\pi a^{-\pi}, \quad (19)$$

with

$$H_\pi(a) = H_0^{(\pi)} E_\pi(a). \quad (20)$$

The acoustic projection is derived from the native branch through

$$p_{\text{ac}} = 3 - \frac{\pi}{100} = 2.968584073464, \quad (21)$$

so that

$$\mathcal{G}_T = \frac{p_{\text{ac}}}{p_C} = 0.9449296585513721, \quad (22)$$

and

$$\sqrt{\mathcal{G}_T} = 0.9720749243506758. \quad (23)$$

Define the nonconstant native contribution

$$X_\pi(a) \equiv E_\pi(a)^2 - \mathcal{B}_\pi = \mathcal{T}_\pi a^{-\pi}. \quad (24)$$

The exact normalized acoustic projector is

$$X_{\text{ac}}(a) = \mathcal{T}_\pi^{1-\mathcal{G}_T} X_\pi(a)^{\mathcal{G}_T} = \mathcal{T}_\pi a^{-p_{\text{ac}}}, \quad (25)$$

which preserves the native normalization at $a = 1$. Hence

$$E_{\text{ac}}(a)^2 = \mathcal{B}_\pi + \mathcal{T}_\pi a^{-p_{\text{ac}}}, \quad (26)$$

and the acoustic comparison normalization is

$$H_{0,\text{ac}}^{\text{CPTG}} = H_0^{(\pi)} \sqrt{\mathcal{G}_T} = 67.4777967351 \text{ km s}^{-1} \text{ Mpc}^{-1}. \quad (27)$$

This acoustic normalization is the CPTG CMB-facing comparison coordinate. These relations follow from the registered Pi-branch acoustic construction and are carried into the direct CMB-to-DESI standard-ruler comparison [5, 6]. They are fixed before the Planck or SH0ES endpoint comparisons are made.

This dependency chain is important because neither $H_0^{(\pi)}$, $\sqrt{\mathcal{G}_T}$, nor the affine luminosity factor $\sqrt{10/9}$ is obtained from the Planck or SH0ES endpoint values. The observational values enter only after the native, acoustic, and luminosity quantities have been fixed.

3 Comparison Coordinates and Observational Status

The Hubble bridge contains native CPTG quantities, derived CPTG comparison coordinates, and external observational coordinates. The categories are distinct but are used together in the bridge.

The compact present-epoch bridge is

$$67.4778 \longleftarrow 69.4163 \longrightarrow 73.1712. \quad (28)$$

The corresponding native-relative endpoint factors are

$$\frac{H_{0,\text{ac}}^{\text{CPTG}}}{H_0^{(\pi)}} = 0.972074924351, \quad \frac{H_{0,L}^{\text{aff}}}{H_0^{(\pi)}} = 1.054092553389460, \quad (29)$$

while the acoustic-to-luminosity endpoint separation is

$$\frac{H_{0,L}^{\text{aff}}}{H_{0,\text{ac}}^{\text{CPTG}}} - 1 = 0.0843737730 = 8.43737730\%. \quad (30)$$

Quantity	Value	Class	Role in the bridge
$a_{\pi,0}$	$5.568327996831708 \times 10^{-10} \text{ m s}^{-2}$	Native invariant	Geometric acceleration anchor upstream of the observational comparison coordinates.
$H_0^{(\pi)}$	69.4162507897	Derived branch quantity	Derived from $a_{\pi,0}$ and $\mathcal{B}_\pi$; not set by Planck or SH0ES.
$\sqrt{\mathcal{G}_T}$	0.9720749243506758	Derived branch quantity	Maps the native branch to the acoustic comparison coordinate.
$H_{0,\text{ac}}^{\text{CPTG}}$	67.4777967351	Derived comparison coordinate	Fixed acoustic/CMB projection compared with the Planck base- Λ CDM inference.
$\mathcal{G}_L^{\text{aff}}$	10/9	Derived comparison coordinate	Registered affine luminosity normalization.
Γ_L^{aff}	1.054092553389460	Derived comparison coordinate	Fixed luminosity rate factor $\sqrt{10/9}$.
Ξ_L^{aff}	0.948683298050514	Derived comparison coordinate	Fixed reciprocal luminosity clock factor $\sqrt{9/10}$.
$H_{0,L}^{\text{aff}}$	73.1711530416	Derived comparison coordinate	Fixed affine luminosity endpoint compared with the local distance ladder.
H_0^{Planck}	67.4 ± 0.5	External base- Λ CDM CMB inference	Planck 2018 model-inferred CMB/acoustic comparison value.
H_0^{SH0ES}	73.17 ± 0.86	External local distance-ladder inference	Independent four-anchor validation coordinate for the luminosity endpoint.

Table 1: Native, projected, and observational coordinates for the Hubble-tension bridge. H_0 values are in $\text{km s}^{-1} \text{Mpc}^{-1}$. The CPTG acoustic and affine luminosity entries are fixed theory-side comparison coordinates; quoted observational uncertainties belong to Planck and SH0ES rather than to those CPTG coordinates.

The left and right endpoints are independently fixed projections of the native branch. Planck and SH0ES are subsequently compared with those projected coordinates rather than used to define them.

4 Acoustic Projection and Planck Comparison

The acoustic comparison is the geometry-fixed side of the bridge. Using only the fixed CPTG quantities,

$$H_{0,\text{ac}}^{\text{CPTG}} = 69.4162507897 \times 0.9720749243506758 = 67.4777967351 \text{ km s}^{-1} \text{Mpc}^{-1}. \quad (31)$$

Relative to the Planck 2018 base- Λ CDM CMB inference,

$$\Delta H_{0,\text{ac}} = H_{0,\text{ac}}^{\text{CPTG}} - H_0^{\text{Planck}} = 0.0777967351 \text{ km s}^{-1} \text{Mpc}^{-1}. \quad (32)$$

If the fixed CPTG quantities are treated as exact theory coordinates for this comparison and only the quoted Planck uncertainty is used, the residual is

$$\frac{\Delta H_{0,\text{ac}}}{0.5} \simeq 0.156\sigma. \quad (33)$$

No additional theory uncertainty is assigned in that quoted residual. If an uncertainty

distribution is assigned to the native CPTG inputs, it must be propagated before the residual is interpreted as a combined statistical pull.

The acoustic result is therefore a direct geometry-to-observation comparison: the fixed native branch and exact acoustic projection produce an H_0 -like comparison coordinate that can be compared with the Planck base- Λ CDM inference. The same acoustic construction is used in the current Pi-branch and direct CMB-to-DESI standard-ruler papers [5, 6].

5 Affine Luminosity Projection and SH0ES Validation

The luminosity side of the bridge is now fixed by the registered affine CPTG comparison map rather than by conditioning the branch normalization on SH0ES. In the current Pi-branch construction, three theory-side routes close on the same one-ninth response increment: the matched polarization response $C_{\Pi,\star} = I/9$, the structural ratio $C_H/C_R = 1/9$, and the acoustic rank quotient $\mathcal{R}_3 = \Omega_{m,\text{ac}}/\mathcal{G}_T = 1/3$, whose square is $1/9$ [5]. The registered affine normalization is therefore

$$\mathcal{G}_L^{\text{aff}} = 1 + \mathcal{R}_3^2 = 1 + \frac{1}{9} = \frac{10}{9}. \quad (34)$$

The corresponding rate and reciprocal clock factors are

$$\Gamma_L^{\text{aff}} = \sqrt{\mathcal{G}_L^{\text{aff}}} = \sqrt{\frac{10}{9}} = 1.054092553389460, \quad (35)$$

$$\Xi_L^{\text{aff}} = (\Gamma_L^{\text{aff}})^{-1} = \sqrt{\frac{9}{10}} = 0.948683298050514. \quad (36)$$

The matched-normalization luminosity trajectory is

$$H_L^{\text{aff}}(a) = \Gamma_L^{\text{aff}} H_\pi(a) = \sqrt{\frac{10}{9}} H_\pi(a). \quad (37)$$

Because the same constant multiplies both the full history and its present normalization,

$$E_L^{\text{aff}}(a) \equiv \frac{H_L^{\text{aff}}(a)}{H_{0,L}^{\text{aff}}} = E_\pi(a) \quad (38)$$

exactly. The luminosity projection therefore preserves the normalized native shape while changing the comparison normalization.

At the present phase,

$$H_{0,L}^{\text{aff}} = H_0^{(\pi)} \sqrt{\frac{10}{9}} = 73.1711530416 \text{ km s}^{-1} \text{ Mpc}^{-1}. \quad (39)$$

The branch clock follows the same constant map,

$$\frac{dt_L^{\text{aff}}}{dt_\pi} = \Xi_L^{\text{aff}} = \sqrt{\frac{9}{10}}, \quad H_L^{\text{aff}} dt_L^{\text{aff}} = H_\pi dt_\pi = dN, \quad (40)$$

so with the same zero-time convention,

$$t_L^{\text{aff}}(a) = \sqrt{\frac{9}{10}} t_\pi(a), \quad (41)$$

and

$$t_{0,L}^{\text{aff}} = 12.0996216046 \text{ Gyr}. \quad (42)$$

The four-anchor SH0ES observation [2]

$$H_0^{\text{SH0ES}} = 73.17 \pm 0.86 \text{ km s}^{-1} \text{ Mpc}^{-1} \quad (43)$$

is applied only after the affine endpoint has been derived. The central-value residual is

$$\Delta H_{0,L}^{\text{SH0ES}} \equiv H_{0,L}^{\text{aff}} - H_0^{\text{SH0ES}} = 0.0011530416 \text{ km s}^{-1} \text{ Mpc}^{-1}, \quad (44)$$

or approximately

$$\frac{\Delta H_{0,L}^{\text{SH0ES}}}{0.86} \simeq 0.00134\sigma. \quad (45)$$

The quoted SH0ES uncertainty is therefore an observational comparison uncertainty; it is not assigned to $\mathcal{G}_L^{\text{aff}}$, Γ_L^{aff} , Ξ_L^{aff} , or $H_{0,L}^{\text{aff}}$ in this fixed-coordinate comparison.

At fixed dimensionless distance-redshift shape, the corresponding normalization-scale distance-modulus relation is

$$\Delta\mu = 5 \log_{10} \left(\frac{H_{0,\text{ref}}}{H_{0,L}^{\text{aff}}} \right). \quad (46)$$

Using the native branch as the reference gives

$$\Delta\mu_{\pi \rightarrow L} = 5 \log_{10} \left(\frac{69.4162507897}{73.1711530416} \right) = -0.1143937 \text{ mag}, \quad (47)$$

while using the fixed CPTG acoustic projection gives

$$\Delta\mu_{\text{ac} \rightarrow L} = 5 \log_{10} \left(\frac{67.4777967351}{73.1711530416} \right) = -0.1758950 \text{ mag}. \quad (48)$$

If the SH0ES measurement is used as the observational comparison scale, its quoted uncertainty corresponds approximately to

$$\sigma_{\Delta\mu}^{\text{SH0ES}} = \frac{5}{\ln 10} \frac{0.86}{73.17} = 0.0255223 \text{ mag}. \quad (49)$$

This is an uncertainty on the observational comparison, not on the fixed affine luminosity coordinate. The magnitude offsets above remain normalization-scale comparison coordinates and are not asserted to be universal redshift-independent magnitude corrections.

For continuity with the earlier Hubble-bridge construction, the historical data-conditioned scaffold

$$H_{\text{lum}}^{\text{anchor}}(a) = H_0^{\text{SH0ES}} E_{\text{ac},\rho}(a) \quad (50)$$

may still be retained as a separate observational diagnostic [5]. It does not define the affine luminosity branch, its clock, its endpoint, or the primary figures in this paper.

6 Bridge Diagnostics and Comparison Trajectories

Figure 1 displays the evidentiary structure directly. Planck and SH0ES retain their quoted observational uncertainties, while the CPTG acoustic projection, native branch, and affine luminosity projection are fixed theory-side coordinates. The near coincidence of the affine luminosity point and the SH0ES point is therefore a comparison result rather than a construction identity.

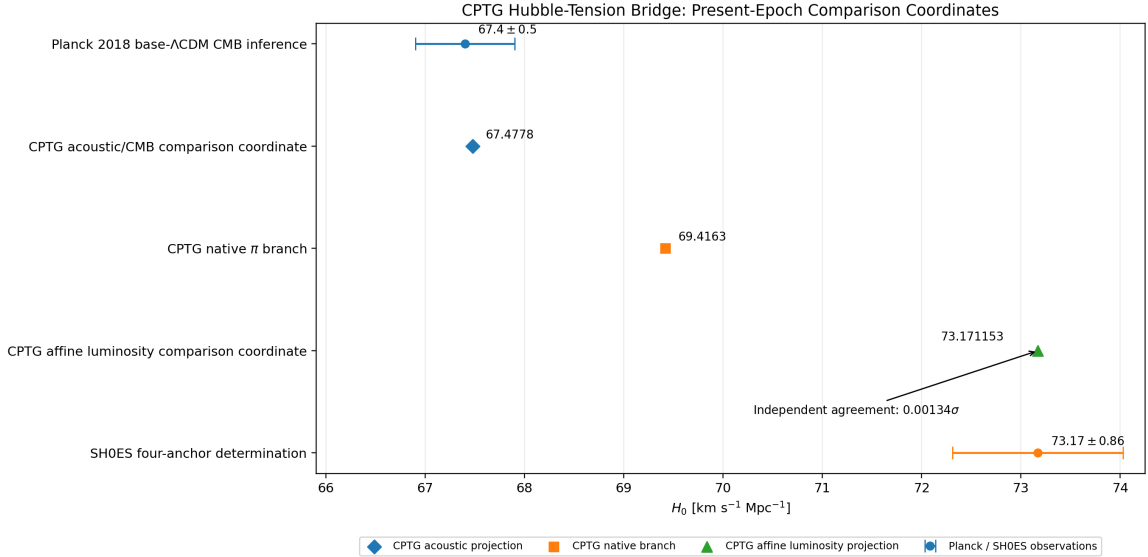


Figure 1: Present-epoch Hubble-bridge map generated from the fixed CPTG native, acoustic, and affine luminosity coordinates together with the declared Planck and four-anchor SH0ES observational values. The acoustic coordinate is compared with Planck, while the independently derived affine luminosity endpoint $H_{0,L}^{\text{aff}} = 73.1711530416 \text{ km s}^{-1} \text{ Mpc}^{-1}$ is compared with SH0ES $73.17 \pm 0.86 \text{ km s}^{-1} \text{ Mpc}^{-1}$. The luminosity theory point carries no SH0ES-derived uncertainty.

A full-history comparison can be constructed without promoting the local ladder into a measured expansion history. The acoustic/CMB density-weighted diagnostic uses

$$\Omega_{m,\text{ac}} = \frac{p_{\text{ac}}}{3\pi} = 0.314976552850, \quad (51)$$

$$\Omega_{\Lambda,\text{ac}} = 1 - \Omega_{m,\text{ac}} = 0.685023447150. \quad (52)$$

The corresponding acoustic-density comparison background is

$$E_{\text{ac},\rho}(a)^2 = \Omega_{\Lambda,\text{ac}} + \Omega_{m,\text{ac}} a^{-p_{\text{ac}}}, \quad (53)$$

so that

$$H_{\text{ac},\rho}(a) = H_{0,\text{ac}}^{\text{CPTG}} E_{\text{ac},\rho}(a). \quad (54)$$

The term “density” identifies the acoustic/CMB density weighting of this comparison background; it does not introduce a separate native expansion branch. The symbols $\Omega_{m,\text{ac}}$ and $\Omega_{\Lambda,\text{ac}}$ are derived CPTG comparison weights fixed by p_{ac} ; they are not a new observational fit of the conventional matter and cosmological-constant density parameters.

The native geometric trajectory remains

$$H_{\pi}(a) = H_0^{(\pi)} \sqrt{\mathcal{B}_{\pi} + \mathcal{T}_{\pi} a^{-\pi}}. \quad (55)$$

The affine luminosity trajectory is instead fixed directly from that native history,

$$H_L^{\text{aff}}(a) = \sqrt{\frac{10}{9}} H_{\pi}(a), \quad (56)$$

with constant clock ratio

$$\frac{dt_L^{\text{aff}}}{dt_{\pi}} = \sqrt{\frac{9}{10}}. \quad (57)$$

All dimensional ages in this section use $1 \text{ Mpc} = 3.0856775814913673 \times 10^{19} \text{ km}$ and the Julian year $1 \text{ yr} = 31,557,600 \text{ s}$.

For each displayed trajectory,

$$t(a) = \int_0^a \frac{da'}{a' H(a')}. \quad (58)$$

The acoustic-density present-age coordinate is therefore

$$t_{0,\text{ac},\rho}^{\text{CPTG}} = \frac{1}{H_{0,\text{ac}}^{\text{CPTG}}} \int_0^1 \frac{da}{a E_{\text{ac},\rho}(a)} \quad (59)$$

$$= \frac{2}{p_{\text{ac}} H_{0,\text{ac}}^{\text{CPTG}} \sqrt{\Omega_{\Lambda,\text{ac}}}} \sinh^{-1} \sqrt{\frac{\Omega_{\Lambda,\text{ac}}}{\Omega_{m,\text{ac}}}} \quad (60)$$

$$= 13.926455176722689 \text{ Gyr}. \quad (61)$$

The native geometric age remains

$$t_{0,\pi} = 12.7541210323 \text{ Gyr}, \quad (62)$$

while the affine luminosity comparison clock is

$$t_{0,L}^{\text{aff}} = \sqrt{\frac{9}{10}} t_{0,\pi} = 12.0996216046 \text{ Gyr}. \quad (63)$$

For clarity, the exact acoustic branch is distinct from the acoustic-density diagnostic. The exact acoustic projected background is

$$E_{\text{ac}}(a)^2 = \mathcal{B}_\pi + \mathcal{T}_\pi a^{-p_{\text{ac}}}, \quad (64)$$

with present acoustic-branch age

$$t_{0,\text{ac}}^{\text{CPTG}} = 13.88517358 \text{ Gyr}. \quad (65)$$

The exact acoustic branch is the physical projection defined earlier; $E_{\text{ac},\rho}$ is a separate density-weighted comparison diagnostic and does not define another branch clock.

The affine luminosity and native rates do not cross at any finite common phase because

$$\frac{H_L^{\text{aff}}(a)}{H_\pi(a)} = \sqrt{\frac{10}{9}} > 1 \quad (66)$$

for the entire trajectory. Their clock relation is likewise an exact constant rescaling,

$$t_L^{\text{aff}}(a) - t_\pi(a) = \left(\sqrt{\frac{9}{10}} - 1 \right) t_\pi(a), \quad (67)$$

so the luminosity clock remains below the native clock at equal phase. At the present phase,

$$t_{0,L}^{\text{aff}} - t_{0,\pi} = -0.6544994277 \text{ Gyr}. \quad (68)$$

The magnitude of this separation grows monotonically from zero under the common zero-time convention. The integrated values $t_{0,\pi}$, $t_{0,\text{ac}}^{\text{CPTG}}$, $t_{0,\text{ac},\rho}^{\text{CPTG}}$, and $t_{0,L}^{\text{aff}}$ are time coordinates of the declared native, projected, or diagnostic backgrounds; they are not independent observational determinations of the physical age of the Universe.

The trajectory figure visualizes the rate and clock consequences of the declared comparison forms; it does not supply an additional observational validation.

7 Interpretation and External Cross-Checks

The bridge is a comparison-coordinate formulation. Planck and SH0ES are external observational coordinates with their quoted uncertainties, while the native CPTG branch supplies the common geometric reference. The acoustic and affine luminosity projections are fixed from CPTG before those observations are applied:

$$H_{0,\text{ac}}^{\text{CPTG}} = H_0^{(\pi)} \sqrt{\mathcal{G}_T}, \quad H_{0,L}^{\text{aff}} = H_0^{(\pi)} \sqrt{\frac{10}{9}}. \quad (69)$$

The Planck and SH0ES values therefore test different fixed projections of the same native history. In particular, the SH0ES result is not used to normalize the affine luminosity branch; its agreement with $H_{0,L}^{\text{aff}}$ is an external validation of that theory-side endpoint.

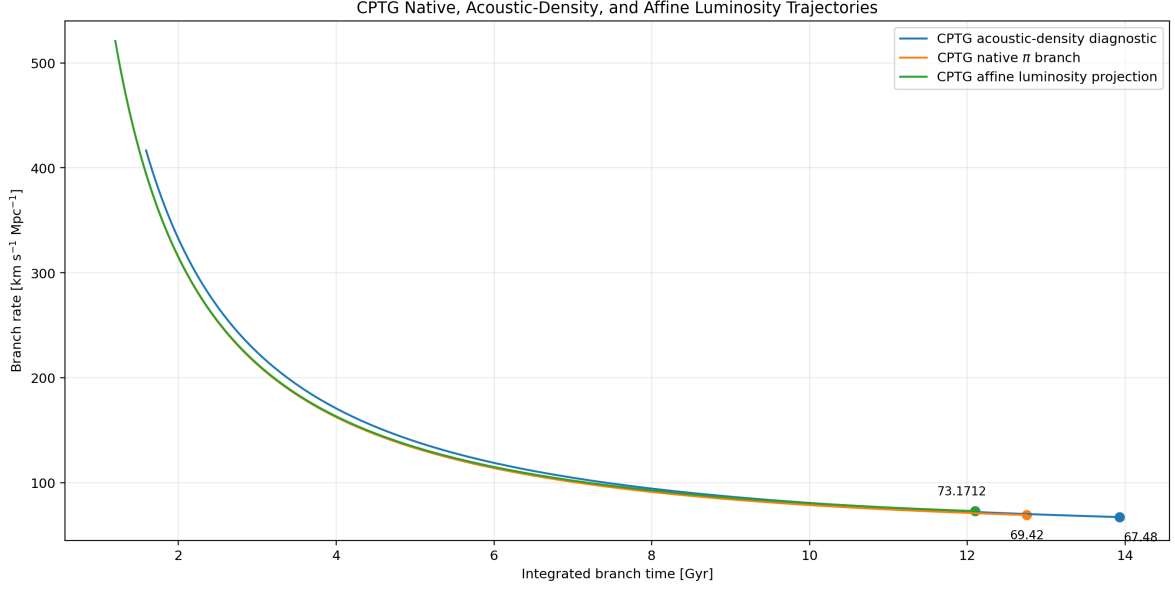


Figure 2: Native and comparison-layer trajectories generated from the equations stated in the text. The acoustic-density curve is the diagnostic $H_{ac,\rho}(a)$; the native curve is $H_{\pi}(a)$; and the affine luminosity curve is $H_L^{\text{aff}}(a) = \sqrt{10/9} H_{\pi}(a)$. All three use the same scale-factor sequence, but each curve is plotted against its own integrated branch time $t_b(a)$, so equal horizontal positions do not denote equal geometric phase. At equal phase the affine luminosity rate is always the fixed factor $\sqrt{10/9}$ above the native rate, and there is no finite-phase native–luminosity rate crossing.

Independent distance channels provide useful external cross-checks of the observational calibration landscape without being prerequisites for the CPTG bridge. Relevant routes include geometric maser distances, alternative standard-candle calibrations such as the tip of the red-giant branch, strong-lens time-delay distances, and controlled splits of the Cepheid–SN calibration chain by host environment, wavelength, or calibration rung. The SH0ES ladder is particularly informative because its simultaneous fit links geometric anchors, Cepheid calibration, calibrated SNe Ia, and Hubble-flow SNe Ia with their covariance structure [3].

The 2025 Chicago–Carnegie Hubble Program analysis uses independent Cepheid, TRGB, and JAGB distance routes, while the 2024 JWST analysis by Riess et al. compares Cepheid, TRGB, and JAGB subsamples after explicitly treating sample selection and common-supernova covariance [7, 8]. The CCHP paper was subsequently corrected by an erratum that revised an Appendix B slope significance but stated that the correction does not affect the rest of the Hubble analysis [9]. These channels are therefore retained as external calibration cross-checks rather than as alternate normalizations imposed on the bridge.

The comparison objective is to keep theory construction and observational validation in the correct order: the native, acoustic, and affine luminosity coordinates are fixed first, and external distance measurements are then used to test the corresponding projected coordinate.

8 Conclusion

The CPTG Hubble bridge represents the Planck base- Λ CDM CMB inference and the four-anchor SH0ES distance-ladder determination as external observational tests of two fixed projections of one native geometric history. The shared phase is $N = \ln a$, while the native, acoustic, and luminosity branches accumulate their own comparison-clock readings through

$$H_b dt_b = dN. \quad (70)$$

The native normalization

$$H_0^{(\pi)} = 69.4162507897 \text{ km s}^{-1} \text{ Mpc}^{-1} \quad (71)$$

follows from the CPTG acceleration anchor and branch weight. The exact acoustic projector gives

$$H_{0,\text{ac}}^{\text{CPTG}} = 67.4777967351 \text{ km s}^{-1} \text{ Mpc}^{-1}, \quad (72)$$

which lies 0.156σ from the Planck 2018 base- Λ CDM inference 67.4 ± 0.5 when the CPTG coordinate is treated as fixed.

Independently, the registered affine luminosity normalization

$$\mathcal{G}_L^{\text{aff}} = \frac{10}{9} \quad (73)$$

gives

$$H_{0,L}^{\text{aff}} = 73.1711530416 \text{ km s}^{-1} \text{ Mpc}^{-1}. \quad (74)$$

The four-anchor SH0ES determination

$$H_0^{\text{SH0ES}} = 73.17 \pm 0.86 \text{ km s}^{-1} \text{ Mpc}^{-1} \quad (75)$$

is not used to define that endpoint. The difference is

$$H_{0,L}^{\text{aff}} - H_0^{\text{SH0ES}} = 0.0011530416 \text{ km s}^{-1} \text{ Mpc}^{-1} \simeq 0.00134\sigma. \quad (76)$$

The theory-constructed acoustic and luminosity endpoints are separated by

$$\frac{H_{0,L}^{\text{aff}}}{H_{0,\text{ac}}^{\text{CPTG}}} - 1 = 8.43737730\%. \quad (77)$$

The quoted Planck and SH0ES observational endpoints themselves remain separated by 5.80σ under the stated independent-error arithmetic. CPTG does not discard either result or interpolate between them to define the native branch. Instead, the acoustic and luminosity sides are derived independently from the same native geometry and then compared with Planck and SH0ES, respectively.

The updated luminosity clock also changes the full-history interpretation. Because

$$H_L^{\text{aff}}(a) = \sqrt{\frac{10}{9}} H_\pi(a), \quad t_L^{\text{aff}}(a) = \sqrt{\frac{9}{10}} t_\pi(a), \quad (78)$$

the normalized luminosity shape is exactly the native shape, the luminosity and native rates have no finite-phase crossing, and the clock relation is a constant rescaling rather than the phase-dependent behavior of the earlier SH0ES-conditioned scaffold. At the present phase, $t_{0,L}^{\text{aff}} = 12.0996216046$ Gyr and $t_{0,L}^{\text{aff}} - t_{0,\pi} = -0.6544994277$ Gyr. The Hubble bridge is therefore a projection-and-clock statement built on one native branch, with Planck and SH0ES serving as external tests rather than branch-defining inputs.

References

- [1] Planck Collaboration, *Planck 2018 results. VI. Cosmological parameters*, doi:10.1051/0004-6361/201833910.
- [2] L. Breuval et al., *Small Magellanic Cloud Cepheids Observed with the Hubble Space Telescope Provide a New Anchor for the SH0ES Distance Ladder*, *Astrophysical Journal* **973**, 30 (2024), doi:10.3847/1538-4357/ad630e.
- [3] A. G. Riess et al., *A Comprehensive Measurement of the Local Value of the Hubble Constant with 1 km/s/Mpc Uncertainty from the Hubble Space Telescope and the SH0ES Team*, doi:10.3847/2041-8213/ac5c5b.
- [4] Carter L. Glass Jr., *Curvature Polarization Transport Gravity: A Unified Geometric Framework for Structure, Cosmology, and Nuclear Reactions*, Zenodo (2026), DOI: 10.5281/zenodo.23172626.
- [5] Carter L. Glass Jr., *CPTG Geometric Pi Branch: Cosmological Comparison Coordinates, Version II*, Zenodo (2026), DOI: 10.5281/zenodo.22907311.
- [6] Carter L. Glass Jr., *CPTG and the Cosmic Standard Ruler: A Direct CMB-to-DESI Test of the Acoustic Projection*, Zenodo (2026), DOI: 10.5281/zenodo.22935182.
- [7] W. L. Freedman, B. F. Madore, T. J. Hoyt, I. S. Jang, A. J. Lee, and K. A. Owens, *Status Report on the Chicago-Carnegie Hubble Program (CCHP): Measurement of the Hubble Constant Using the Hubble and James Webb Space Telescopes*, *Astrophysical Journal* **985**, 203 (2025), doi:10.3847/1538-4357/adce78.
- [8] A. G. Riess et al., *JWST Validates HST Distance Measurements: Selection of Supernova Subsample Explains Differences in JWST Estimates of Local H_0* , *Astrophysical Journal* **977**, 120 (2024), doi:10.3847/1538-4357/ad8c21.
- [9] W. L. Freedman, B. F. Madore, T. J. Hoyt, I. S. Jang, A. J. Lee, and K. A. Owens, *Erratum: “Status Report on the Chicago–Carnegie Hubble Program (CCHP): Measurement of the Hubble Constant Using the Hubble and James Webb Space Telescopes”*, *Astrophysical Journal* **993**, 252 (2025), doi:10.3847/1538-4357/ae146d.